

Rationalising the Denominator

Worked Solutions with TikZ Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref a6)

This document contains all 59 *Rationalising the Denominator* questions, each with a fully worked solution and a TikZ diagram. The whole topic rests on one idea: **multiply by a cleverly chosen form of 1**, so the diagrams reflect that:

- $\frac{\sqrt{a}}{\sqrt{a}}$ **cards** for a single-surd denominator;
- **conjugate cards** for a two-term denominator (the difference of two squares clears the surd);
- **solution-flow** diagrams for simplifying, combining, and solving equations;
- a **rectangle** diagram for the area problem.

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Rationalising the Denominator 01 (a6-001)

Difficulty: Foundation / Marks: 1

Rationalise the denominator: $\frac{1}{\sqrt{2}}$

Worked solution.

Multiply the top and bottom of the fraction by $\sqrt{2}$.

$$\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Multiplying by $\frac{\sqrt{2}}{\sqrt{2}}$ does not change the value of the fraction (it equals 1), but it removes the surd from the denominator.

Final answer: $\frac{\sqrt{2}}{2}$

Diagram.

$\frac{\sqrt{2}}{\sqrt{2}} = 1$

 $\frac{1}{\sqrt{2}} \xrightarrow[\text{multiply top and bottom by } \sqrt{2}]{\text{denominator: } \sqrt{2} \times \sqrt{2} = 2} = \frac{\sqrt{2}}{2}$

Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 02 (a6-002)

Difficulty: Foundation / Marks: 1

Rationalise the denominator: $\frac{1}{\sqrt{5}}$

Worked solution.

Multiply the top and bottom by $\sqrt{5}$.

$$\frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

$\sqrt{5} \times \sqrt{5} = 5$ on the bottom, so the surd disappears.

Final answer: $\frac{\sqrt{5}}{5}$

Diagram.

$\frac{\sqrt{5}}{\sqrt{5}} = 1$

 $\frac{1}{\sqrt{5}} \xrightarrow[\text{multiply top and bottom by } \sqrt{5}]{\text{denominator: } \sqrt{5} \times \sqrt{5} = 5} = \frac{\sqrt{5}}{5}$

Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 03 (a6-003)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{8}{\sqrt{2}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{2}$.

$$\frac{8}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{8\sqrt{2}}{2}$$

$$\sqrt{2} \times \sqrt{2} = 2.$$

Step 2: Simplify the fraction.

$$\frac{8\sqrt{2}}{2} = 4\sqrt{2}$$

Divide top and bottom by 2.

Final answer: $4\sqrt{2}$

Diagram.

$\frac{\sqrt{2}}{\sqrt{2}} = 1$

$$\frac{8}{\sqrt{2}} \xrightarrow[\text{denominator: } \sqrt{2} \times \sqrt{2} = 2]{\text{multiply top and bottom by } \sqrt{2}} = 4\sqrt{2}$$

Then $\frac{8\sqrt{2}}{2} = 4\sqrt{2}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 04 (a6-004)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{6}{\sqrt{3}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{3}$.

$$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3}$$

$$\sqrt{3} \times \sqrt{3} = 3.$$

Step 2: Simplify the fraction.

$$\frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

Divide top and bottom by 3.

Final answer: $2\sqrt{3}$

Diagram.

$$\frac{\sqrt{3}}{\sqrt{3}} = 1$$

$$\frac{6}{\sqrt{3}} \xrightarrow[\text{denominator: } \sqrt{3} \times \sqrt{3} = 3]{\text{multiply top and bottom by } \sqrt{3}} = 2\sqrt{3}$$

Then $\frac{6\sqrt{3}}{3} = 2\sqrt{3}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 05 (a6-005)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{14}{\sqrt{7}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{7}$.

$$\frac{14}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} = \frac{14\sqrt{7}}{7}$$

$$\sqrt{7} \times \sqrt{7} = 7.$$

Step 2: Simplify the fraction.

$$\frac{14\sqrt{7}}{7} = 2\sqrt{7}$$

$$14 \div 7 = 2.$$

Final answer: $2\sqrt{7}$

Diagram.

$$\frac{\sqrt{7}}{\sqrt{7}} = 1$$

$$\frac{14}{\sqrt{7}} \xrightarrow[\text{denominator: } \sqrt{7} \times \sqrt{7} = 7]{\text{multiply top and bottom by } \sqrt{7}} = 2\sqrt{7}$$

Then $\frac{14\sqrt{7}}{7} = 2\sqrt{7}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 06 (a6-006)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{20}{\sqrt{5}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{5}$.

$$\frac{20}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{20\sqrt{5}}{5}$$

$$\sqrt{5} \times \sqrt{5} = 5.$$

Step 2: Simplify the fraction.

$$\frac{20\sqrt{5}}{5} = 4\sqrt{5}$$

$$20 \div 5 = 4.$$

Final answer: $4\sqrt{5}$

Diagram.

$$\frac{20}{\sqrt{5}} \xrightarrow[\text{denominator: } \sqrt{5} \times \sqrt{5} = 5]{\begin{array}{c} \boxed{\frac{\sqrt{5}}{\sqrt{5}} = 1} \\ \text{multiply top and bottom by } \sqrt{5} \end{array}} = 4\sqrt{5}$$

Then $\frac{20\sqrt{5}}{5} = 4\sqrt{5}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 07 (a6-007)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{9}{\sqrt{3}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{3}$.

$$\frac{9}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{3}}{3}$$

$$\sqrt{3} \times \sqrt{3} = 3.$$

Step 2: Simplify the fraction.

$$\frac{9\sqrt{3}}{3} = 3\sqrt{3}$$

$$9 \div 3 = 3.$$

Final answer: $3\sqrt{3}$

Diagram.

$$\frac{9}{\sqrt{3}} \xrightarrow[\text{denominator: } \sqrt{3} \times \sqrt{3} = 3]{\begin{array}{c} \boxed{\frac{\sqrt{3}}{\sqrt{3}} = 1} \\ \text{multiply top and bottom by } \sqrt{3} \end{array}} = 3\sqrt{3}$$

Then $\frac{9\sqrt{3}}{3} = 3\sqrt{3}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 08 (a6-008)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{10}{\sqrt{2}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{2}$.

$$\frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\sqrt{2}}{2}$$

$$\sqrt{2} \times \sqrt{2} = 2.$$

Step 2: Simplify.

$$5\sqrt{2}$$

$$10 \div 2 = 5.$$

Final answer: $5\sqrt{2}$

Diagram.

$$\frac{10}{\sqrt{2}} \xrightarrow[\text{denominator: } \sqrt{2} \times \sqrt{2} = 2]{\begin{array}{c} \boxed{\frac{\sqrt{2}}{\sqrt{2}} = 1} \\ \text{multiply top and bottom by } \sqrt{2} \end{array}} = 5\sqrt{2}$$

Then $\frac{10\sqrt{2}}{2} = 5\sqrt{2}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 09 (a6-009)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{24}{\sqrt{6}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{6}$.

$$\frac{24\sqrt{6}}{6}$$

$$\sqrt{6} \times \sqrt{6} = 6.$$

Step 2: Simplify.

$$4\sqrt{6}$$

$$24 \div 6 = 4.$$

Final answer: $4\sqrt{6}$

Diagram.

$$\frac{\sqrt{6}}{\sqrt{6}} = 1$$

$$\frac{24}{\sqrt{6}} \xrightarrow[\text{denominator: } \sqrt{6} \times \sqrt{6} = 6]{\text{multiply top and bottom by } \sqrt{6}} = 4\sqrt{6}$$

Then $\frac{24\sqrt{6}}{6} = 4\sqrt{6}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 10 (a6-010)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{15}{\sqrt{5}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{5}$.

$$\frac{15\sqrt{5}}{5}$$

$$\sqrt{5} \times \sqrt{5} = 5.$$

Step 2: Simplify.

$$3\sqrt{5}$$

$$15 \div 5 = 3.$$

Final answer: $3\sqrt{5}$

Diagram.

$$\frac{\sqrt{5}}{\sqrt{5}} = 1$$

$$\frac{15}{\sqrt{5}} \xrightarrow[\text{denominator: } \sqrt{5} \times \sqrt{5} = 5]{\text{multiply top and bottom by } \sqrt{5}} = 3\sqrt{5}$$

Then $\frac{15\sqrt{5}}{5} = 3\sqrt{5}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 11 (a6-011)

Difficulty: Foundation / Marks: 3

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\sqrt{32} + \frac{10}{\sqrt{2}}$

Worked solution.

Step 1: Simplify $\sqrt{32}$.

$$\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$$

16 is the largest square factor of 32.

Step 2: Rationalise the second term.

$$\frac{10}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}$$

Multiply top and bottom by $\sqrt{2}$, then simplify.

Step 3: Add the like surds.

$$4\sqrt{2} + 5\sqrt{2} = 9\sqrt{2}$$

$$4 + 5 = 9.$$

Final answer: $9\sqrt{2}$

Diagram.

$$\sqrt{32} + \frac{10}{\sqrt{2}} \xrightarrow{\text{simplify and rationalise}} 4\sqrt{2} + 5\sqrt{2} \xrightarrow{\text{collect like surds}} 9\sqrt{2}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 12 (a6-012)

Difficulty: Foundation / Marks: 3

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\sqrt{75} - \frac{6}{\sqrt{3}}$

Worked solution.

Step 1: Simplify $\sqrt{75}$.

$$\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$$

25 is the largest square factor of 75.

Step 2: Rationalise the second term.

$$\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

Multiply top and bottom by $\sqrt{3}$.

Step 3: Subtract the like surds.

$$5\sqrt{3} - 2\sqrt{3} = 3\sqrt{3}$$

$$5 - 2 = 3.$$

Final answer: $3\sqrt{3}$

Diagram.

$$\sqrt{75} - \frac{6}{\sqrt{3}} \xrightarrow{\text{6 simplify and rationalise}} 5\sqrt{3} - 2\sqrt{3} \xrightarrow{\text{collect like surds}} 3\sqrt{3}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 13 (a6-013)

Difficulty: Foundation / Marks: 2

Rationalise the denominator: $\frac{3}{\sqrt{6}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{6}$.

$$\frac{3}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{3\sqrt{6}}{6}$$

$$\sqrt{6} \times \sqrt{6} = 6.$$

Step 2: Simplify the fraction.

$$\frac{\sqrt{6}}{2}$$

Divide top and bottom by 3.

Final answer: $\frac{\sqrt{6}}{2}$

Diagram.

$$\frac{\sqrt{6}}{\sqrt{6}} = 1$$

$$\frac{3}{\sqrt{6}} \xrightarrow[\text{denominator: } \sqrt{6} \times \sqrt{6} = 6]{\text{multiply top and bottom by } \sqrt{6}} = \frac{\sqrt{6}}{2}$$

Then $\frac{3\sqrt{6}}{6} = \frac{\sqrt{6}}{2}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 14 (a6-014)

Difficulty: Foundation / Marks: 2

Rationalise the denominator: $\frac{4}{\sqrt{10}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{10}$.

$$\frac{4\sqrt{10}}{10}$$

$$\sqrt{10} \times \sqrt{10} = 10.$$

Step 2: Simplify by dividing top and bottom by 2.

$$\frac{2\sqrt{10}}{5}$$

The 4 and the 10 share a common factor of 2.

Final answer: $\frac{2\sqrt{10}}{5}$

Diagram.

$$\frac{4}{\sqrt{10}} \xrightarrow[\text{denominator: } \sqrt{10} \times \sqrt{10} = 10]{\text{multiply top and bottom by } \sqrt{10}} = \frac{2\sqrt{10}}{5}$$

$$\frac{\sqrt{10}}{\sqrt{10}} = 1$$

Then $\frac{4\sqrt{10}}{10} = \frac{2\sqrt{10}}{5}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 15 (a6-015)

Difficulty: Foundation / Marks: 3

Rationalise the denominator: $\frac{\sqrt{3}}{\sqrt{8}}$

Worked solution.

Step 1: Simplify $\sqrt{8}$ first.

$$\frac{\sqrt{3}}{\sqrt{8}} = \frac{\sqrt{3}}{2\sqrt{2}}$$

$$\sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}.$$

Step 2: Multiply the top and bottom by $\sqrt{2}$.

$$\frac{\sqrt{3} \times \sqrt{2}}{2\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{6}}{4}$$

$$\sqrt{3} \times \sqrt{2} = \sqrt{6} \text{ and } 2\sqrt{2} \times \sqrt{2} = 2 \times 2 = 4.$$

Final answer: $\frac{\sqrt{6}}{4}$

Diagram.

$$\frac{\sqrt{3}}{\sqrt{8}} \xrightarrow{\sqrt{8} = 2\sqrt{2}} \frac{\sqrt{3}}{2\sqrt{2}} \xrightarrow{\times \frac{\sqrt{2}}{\sqrt{2}}} \frac{\sqrt{6}}{4}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 16 (a6-016)

Difficulty: Foundation / Marks: 2

Rationalise the denominator: $\frac{5}{2\sqrt{3}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{3}$.

$$\frac{5}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{5\sqrt{3}}{2 \times 3}$$

$$2\sqrt{3} \times \sqrt{3} = 2 \times 3 = 6.$$

Step 2: Write the final answer.

$$\frac{5\sqrt{3}}{6}$$

$\frac{5\sqrt{3}}{6}$ cannot be simplified further.

Final answer: $\frac{5\sqrt{3}}{6}$

Diagram.

$$\frac{5}{2\sqrt{3}} \xrightarrow[\text{denominator: } 2\sqrt{3} \times \sqrt{3} = 6]{\begin{array}{c} \boxed{\frac{\sqrt{3}}{\sqrt{3}} = 1} \\ \text{multiply top and bottom by } \sqrt{3} \end{array}} = \frac{5\sqrt{3}}{6}$$

Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 17 (a6-017)

Difficulty: Foundation / Marks: 2

Rationalise the denominator and simplify: $\frac{\sqrt{18}}{\sqrt{2}}$

Worked solution.

Step 1: Use the rule $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$.

$$\sqrt{\frac{18}{2}} = \sqrt{9}$$

When dividing surds with the same type, you can combine them under a single root sign.

Step 2: Evaluate.

$$3$$

$\sqrt{9} = 3$. The answer is a whole number — no surds remain.

Final answer: 3

Diagram.

$$\frac{\sqrt{18}}{\sqrt{2}} \xrightarrow{\text{one root}} \sqrt{\frac{18}{2}} = \sqrt{9} \xrightarrow{\text{evaluate}} 3$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 18 (a6-018)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{2}{1+\sqrt{3}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $1 - \sqrt{3}$.

$$\frac{2}{1+\sqrt{3}} \times \frac{1-\sqrt{3}}{1-\sqrt{3}} = \frac{2(1-\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})}$$

The conjugate has the opposite sign between the terms. Multiplying by it removes the surd from the denominator.

Step 2: Expand the denominator using the difference of two squares.

$$\frac{2(1-\sqrt{3})}{1-3} = \frac{2(1-\sqrt{3})}{-2}$$

$$(1+\sqrt{3})(1-\sqrt{3}) = 1-3 = -2.$$

Step 3: Cancel and simplify.

$$-(1-\sqrt{3}) = -1 + \sqrt{3}$$

Dividing by -2 : $\frac{2}{-2} = -1$. So $a = -1$, $b = 1$, $k = 3$.

Final answer: $-1 + \sqrt{3}$

Diagram.

$$\frac{1-\sqrt{3}}{1-\sqrt{3}} = 1 \text{ (conjugate)}$$

$$\frac{2}{1+\sqrt{3}} \xrightarrow[\text{difference of two squares: } (1+\sqrt{3})(1-\sqrt{3}) = 1-3 = -2]{\text{multiply by the conjugate } 1-\sqrt{3}} = -1 + \sqrt{3}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A+B)(A-B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 19 (a6-019)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{8}{\sqrt{5}-1}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $\sqrt{5} + 1$.

$$\frac{8(\sqrt{5} + 1)}{(\sqrt{5} - 1)(\sqrt{5} + 1)}$$

The conjugate of $\sqrt{5} - 1$ is $\sqrt{5} + 1$.

Step 2: Expand the denominator using the difference of two squares.

$$\frac{8(\sqrt{5} + 1)}{5 - 1} = \frac{8(\sqrt{5} + 1)}{4}$$

$$(\sqrt{5})^2 - 1^2 = 5 - 1 = 4.$$

Step 3: Divide the numerator by 4.

$$2(\sqrt{5} + 1) = 2 + 2\sqrt{5}$$

Since $8 \div 4 = 2$, we get integer coefficients $a = 2$, $b = 2$, $k = 5$.

Final answer: $2 + 2\sqrt{5}$

Diagram.

$$\frac{\sqrt{5} + 1}{\sqrt{5} + 1} = 1 \text{ (conjugate)}$$

$$\frac{8}{\sqrt{5} - 1} \xrightarrow[\text{multiply by the conjugate } \sqrt{5} + 1]{\text{difference of two squares: } (\sqrt{5} - 1)(\sqrt{5} + 1) = 5 - 1 = 4} = 2 + 2\sqrt{5}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 20 (a6-020)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{7}{3-\sqrt{2}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $3 + \sqrt{2}$.

$$\frac{7(3 + \sqrt{2})}{(3 - \sqrt{2})(3 + \sqrt{2})}$$

The conjugate flips the sign between the terms.

Step 2: Expand the denominator.

$$\frac{7(3 + \sqrt{2})}{9 - 2} = \frac{7(3 + \sqrt{2})}{7}$$

$$3^2 - (\sqrt{2})^2 = 9 - 2 = 7.$$

Step 3: Cancel the common factor of 7.

$$3 + \sqrt{2}$$

The 7 in the numerator and denominator cancel, giving $a = 3$, $b = 1$, $k = 2$.

Final answer: $3 + \sqrt{2}$

Diagram.

$$\frac{3 + \sqrt{2}}{3 + \sqrt{2}} = 1 \quad (\text{conjugate})$$

$$\frac{7}{3 - \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } 3 + \sqrt{2}} = 3 + \sqrt{2}$$

difference of two squares: $(3 - \sqrt{2})(3 + \sqrt{2}) = 9 - 2 = 7$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 21 (a6-021)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{4}{\sqrt{6}-2}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $\sqrt{6} + 2$.

$$\frac{4(\sqrt{6} + 2)}{(\sqrt{6} - 2)(\sqrt{6} + 2)}$$

The conjugate has the opposite sign.

Step 2: Expand the denominator using the difference of two squares.

$$\frac{4(\sqrt{6} + 2)}{6 - 4} = \frac{4(\sqrt{6} + 2)}{2}$$

$$(\sqrt{6})^2 - 2^2 = 6 - 4 = 2.$$

Step 3: Divide and expand.

$$2(\sqrt{6} + 2) = 4 + 2\sqrt{6}$$

$\frac{4}{2} = 2$, then multiply through.

Final answer: $4 + 2\sqrt{6}$

Diagram.

$$\frac{\sqrt{6} + 2}{\sqrt{6} + 2} = 1 \text{ (conjugate)}$$

$$\frac{4}{\sqrt{6} - 2} \xrightarrow{\text{multiply by the conjugate } \sqrt{6} + 2} = 4 + 2\sqrt{6}$$

difference of two squares: $(\sqrt{6} - 2)(\sqrt{6} + 2) = 6 - 4 = 2$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 22 (a6-022)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{14}{4 + \sqrt{2}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $4 - \sqrt{2}$.

$$\frac{14(4 - \sqrt{2})}{(4 + \sqrt{2})(4 - \sqrt{2})}$$

The conjugate flips the middle sign.

Step 2: Expand the denominator.

$$\frac{14(4 - \sqrt{2})}{16 - 2} = \frac{14(4 - \sqrt{2})}{14}$$

$$4^2 - (\sqrt{2})^2 = 16 - 2 = 14.$$

Step 3: Cancel the 14.

$$4 - \sqrt{2}$$

The 14 on top and bottom cancel exactly.

Final answer: $4 - \sqrt{2}$

Diagram.

$$\frac{4 - \sqrt{2}}{4 - \sqrt{2}} = 1 \text{ (conjugate)}$$

$$\frac{14}{4 + \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } 4 - \sqrt{2}} = 4 - \sqrt{2}$$

difference of two squares: $(4 + \sqrt{2})(4 - \sqrt{2}) = 16 - 2 = 14$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 23 (a6-023)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{12}{5-\sqrt{13}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $5 + \sqrt{13}$.

$$\frac{12(5 + \sqrt{13})}{(5 - \sqrt{13})(5 + \sqrt{13})}$$

Flip the sign between the terms.

Step 2: Expand the denominator.

$$\frac{12(5 + \sqrt{13})}{25 - 13} = \frac{12(5 + \sqrt{13})}{12}$$

$$5^2 - (\sqrt{13})^2 = 25 - 13 = 12.$$

Step 3: Cancel the 12.

$$5 + \sqrt{13}$$

The 12s cancel, leaving the numerator bracket.

Final answer: $5 + \sqrt{13}$

Diagram.

$$\frac{5 + \sqrt{13}}{5 + \sqrt{13}} = 1 \text{ (conjugate)}$$

$$\frac{12}{5 - \sqrt{13}} \xrightarrow[\text{multiply by the conjugate } 5 + \sqrt{13}]{\text{difference of two squares: } (5 - \sqrt{13})(5 + \sqrt{13}) = 25 - 13 = 12} = 5 + \sqrt{13}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 24 (a6-024)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{8}{3+\sqrt{5}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $3 - \sqrt{5}$.

$$\frac{8(3 - \sqrt{5})}{(3 + \sqrt{5})(3 - \sqrt{5})}$$

Use the conjugate to eliminate the surd in the denominator.

Step 2: Expand the denominator.

$$\frac{8(3 - \sqrt{5})}{9 - 5} = \frac{8(3 - \sqrt{5})}{4}$$

$$3^2 - (\sqrt{5})^2 = 9 - 5 = 4.$$

Step 3: Cancel and expand.

$$2(3 - \sqrt{5}) = 6 - 2\sqrt{5}$$

$\frac{8}{4} = 2$, then multiply through the bracket.

Final answer: $6 - 2\sqrt{5}$

Diagram.

$$\frac{3 - \sqrt{5}}{3 - \sqrt{5}} = 1 \text{ (conjugate)}$$

$$\frac{8}{3 + \sqrt{5}} \xrightarrow{\text{multiply by the conjugate } 3 - \sqrt{5}} = \frac{6 - 2\sqrt{5}}{\text{difference of two squares: } (3 + \sqrt{5})(3 - \sqrt{5}) = 9 - 5 = 4}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 25 (a6-025)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{25}{6 - \sqrt{11}}$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $6 + \sqrt{11}$.

$$\frac{25(6 + \sqrt{11})}{(6 - \sqrt{11})(6 + \sqrt{11})}$$

Switch the sign in the conjugate.

Step 2: Expand the denominator.

$$\frac{25(6 + \sqrt{11})}{36 - 11} = \frac{25(6 + \sqrt{11})}{25}$$

$$6^2 - (\sqrt{11})^2 = 36 - 11 = 25.$$

Step 3: Cancel the common factor of 25.

$$6 + \sqrt{11}$$

The numerator and denominator both contain 25, which cancel to give $a = 6$, $b = 1$, $k = 11$.

Final answer: $6 + \sqrt{11}$

Diagram.

$$\frac{6 + \sqrt{11}}{6 + \sqrt{11}} = 1 \quad (\text{conjugate})$$

$$\frac{25}{6 - \sqrt{11}} \xrightarrow{\text{multiply by the conjugate } 6 + \sqrt{11}} = 6 + \sqrt{11}$$

difference of two squares: $(6 - \sqrt{11})(6 + \sqrt{11}) = 36 - 11 = 25$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 26 (a6-026)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{\sqrt{3}+1}{\sqrt{3}-1}$$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $\sqrt{3} + 1$.

$$\frac{(\sqrt{3} + 1)(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$$

The conjugate of $\sqrt{3} - 1$ is $\sqrt{3} + 1$.

Step 2: Expand the numerator using $(a + b)^2 = a^2 + 2ab + b^2$.

$$\frac{3 + 2\sqrt{3} + 1}{3 - 1} = \frac{4 + 2\sqrt{3}}{2}$$

Numerator: $(\sqrt{3})^2 + 2\sqrt{3} + 1 = 4 + 2\sqrt{3}$. Denominator: $3 - 1 = 2$.

Step 3: Simplify by dividing each term by 2.

$$2 + \sqrt{3}$$

$$\frac{4}{2} = 2 \text{ and } \frac{2\sqrt{3}}{2} = \sqrt{3}.$$

Final answer: $2 + \sqrt{3}$

Diagram.

$$\frac{\sqrt{3} + 1}{\sqrt{3} + 1} = 1 \quad (\text{conjugate})$$

$$\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \xrightarrow{\text{multiply by the conjugate } \sqrt{3} + 1} = 2 + \sqrt{3}$$

difference of two squares: $(\sqrt{3} - 1)(\sqrt{3} + 1) = 3 - 1 = 2$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 27 (a6-027)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{\sqrt{7}+2}{\sqrt{7}-3}$$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $\sqrt{7} + 3$.

$$\frac{(\sqrt{7}+2)(\sqrt{7}+3)}{(\sqrt{7}-3)(\sqrt{7}+3)}$$

Switch the sign between the terms in the denominator.

Step 2: Expand the numerator using FOIL.

$$\frac{7 + 3\sqrt{7} + 2\sqrt{7} + 6}{7 - 9} = \frac{13 + 5\sqrt{7}}{-2}$$

Numerator: $\sqrt{7} \times \sqrt{7} + 3\sqrt{7} + 2\sqrt{7} + 6 = 7 + 5\sqrt{7} + 6 = 13 + 5\sqrt{7}$. Denominator: $7 - 9 = -2$.

Step 3: Split the fraction to get the required form.

$$-\frac{13}{2} - \frac{5}{2}\sqrt{7}$$

Divide each term in the numerator by -2 .

$$\text{Final answer: } -\frac{13}{2} - \frac{5}{2}\sqrt{7}$$

Diagram.

$$\frac{\sqrt{7}+3}{\sqrt{7}+3} = 1 \text{ (conjugate)}$$

$$\frac{\sqrt{7}+2}{\sqrt{7}-3} \xrightarrow[\text{multiply by the conjugate } \sqrt{7}+3]{\text{difference of two squares: } (\sqrt{7}-3)(\sqrt{7}+3) = 7-9 = -2} -\frac{13}{2} - \frac{5}{2}\sqrt{7}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A+B)(A-B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 28 (a6-028)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{5-\sqrt{2}}{3+\sqrt{2}}$$

Worked solution.

Step 1: Multiply the top and bottom by the conjugate $3 - \sqrt{2}$.

$$\frac{(5-\sqrt{2})(3-\sqrt{2})}{(3+\sqrt{2})(3-\sqrt{2})}$$

Use the conjugate to eliminate the surd from the denominator.

Step 2: Expand the numerator and denominator.

$$\frac{15 - 5\sqrt{2} - 3\sqrt{2} + 2}{9 - 2} = \frac{17 - 8\sqrt{2}}{7}$$

Numerator: $15 - 5\sqrt{2} - 3\sqrt{2} + (\sqrt{2})^2 = 15 + 2 - 8\sqrt{2} = 17 - 8\sqrt{2}$. Denominator: $9 - 2 = 7$.

Step 3: Split into the required form.

$$\frac{17}{7} - \frac{8}{7}\sqrt{2}$$

Divide each term by 7.

Final answer: $\frac{17}{7} - \frac{8}{7}\sqrt{2}$

Diagram.

$$\frac{3 - \sqrt{2}}{3 - \sqrt{2}} = 1 \quad (\text{conjugate})$$

$$\frac{5 - \sqrt{2}}{3 + \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } 3 - \sqrt{2}} \frac{17}{7} - \frac{8}{7}\sqrt{2}$$

difference of two squares: $(3 + \sqrt{2})(3 - \sqrt{2}) = 9 - 2 = 7$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 29 (a6-029)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{2\sqrt{5}+1}{3\sqrt{5}-2}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $3\sqrt{5} + 2$.

$$\frac{(2\sqrt{5} + 1)(3\sqrt{5} + 2)}{(3\sqrt{5} - 2)(3\sqrt{5} + 2)}$$

The conjugate flips the sign between the terms.

Step 2: Expand the numerator and denominator.

$$\frac{6 \times 5 + 4\sqrt{5} + 3\sqrt{5} + 2}{9 \times 5 - 4} = \frac{30 + 7\sqrt{5} + 2}{45 - 4} = \frac{32 + 7\sqrt{5}}{41}$$

Numerator: $2\sqrt{5} \times 3\sqrt{5} = 6 \times 5 = 30$; cross terms: $4\sqrt{5} + 3\sqrt{5} = 7\sqrt{5}$; last: $1 \times 2 = 2$. Denominator: $(3\sqrt{5})^2 - 2^2 = 45 - 4 = 41$.

Step 3: Split into the required form.

$$\frac{32}{41} + \frac{7}{41}\sqrt{5}$$

Divide each term by 41.

Final answer: $\frac{32}{41} + \frac{7}{41}\sqrt{5}$

Diagram.

$$\frac{3\sqrt{5} + 2}{3\sqrt{5} + 2} = 1 \quad (\text{conjugate})$$

$$\frac{2\sqrt{5} + 1}{3\sqrt{5} - 2} \xrightarrow{\text{multiply by the conjugate } 3\sqrt{5} + 2} = \frac{32}{41} + \frac{7}{41}\sqrt{5}$$

difference of two squares: $(3\sqrt{5} - 2)(3\sqrt{5} + 2) = 45 - 4 = 41$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 30 (a6-030)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{\sqrt{2}+3}{2\sqrt{2}-1}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $2\sqrt{2} + 1$.

$$\frac{(\sqrt{2} + 3)(2\sqrt{2} + 1)}{(2\sqrt{2} - 1)(2\sqrt{2} + 1)}$$

Conjugate has opposite sign between the terms.

Step 2: Expand the numerator and denominator.

$$\frac{2 \times 2 + \sqrt{2} + 6\sqrt{2} + 3}{4 \times 2 - 1} = \frac{4 + 7\sqrt{2} + 3}{8 - 1} = \frac{7 + 7\sqrt{2}}{7}$$

Numerator: $\sqrt{2} \times 2\sqrt{2} = 4$; cross terms: $\sqrt{2} + 6\sqrt{2} = 7\sqrt{2}$; last: 3. Denominator: $(2\sqrt{2})^2 - 1 = 8 - 1 = 7$.

Step 3: Simplify.

$$1 + \sqrt{2}$$

Divide each term by 7.

Final answer: $1 + \sqrt{2}$

Diagram.

$$\frac{2\sqrt{2} + 1}{2\sqrt{2} + 1} = 1 \quad (\text{conjugate})$$

$$\frac{\sqrt{2} + 3}{2\sqrt{2} - 1} \xrightarrow{\text{multiply by the conjugate } 2\sqrt{2} + 1} = 1 + \sqrt{2}$$

difference of two squares: $(2\sqrt{2} - 1)(2\sqrt{2} + 1) = 8 - 1 = 7$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A+B)(A-B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 31 (a6-031)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{4-\sqrt{3}}{2+\sqrt{3}}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $2 - \sqrt{3}$.

$$\frac{(4 - \sqrt{3})(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})}$$

Flip the sign in the denominator to form the conjugate.

Step 2: Expand the numerator and denominator.

$$\frac{8 - 4\sqrt{3} - 2\sqrt{3} + 3}{4 - 3} = \frac{11 - 6\sqrt{3}}{1}$$

Numerator: $8 - 4\sqrt{3} - 2\sqrt{3} + (\sqrt{3})^2 = 8 + 3 - 6\sqrt{3} = 11 - 6\sqrt{3}$. Denominator: $4 - 3 = 1$.

Step 3: Write final answer.

$$11 - 6\sqrt{3}$$

Dividing by 1 leaves the numerator unchanged.

Final answer: $11 - 6\sqrt{3}$

Diagram.

$$\frac{2 - \sqrt{3}}{2 - \sqrt{3}} = 1 \text{ (conjugate)}$$

$$\frac{4 - \sqrt{3}}{2 + \sqrt{3}} \xrightarrow{\text{multiply by the conjugate } 2 - \sqrt{3}} 11 - 6\sqrt{3}$$

difference of two squares: $(2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A+B)(A-B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 32 (a6-032)

Difficulty: Foundation / Marks: 4

Express in the form $k(\sqrt{x} \pm \sqrt{y})$, where x and y are integers and k is an integer or fraction:

$$\frac{6}{\sqrt{5} - \sqrt{2}}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{5} + \sqrt{2}$.

$$\frac{6(\sqrt{5} + \sqrt{2})}{(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})}$$

The conjugate of $\sqrt{5} - \sqrt{2}$ is $\sqrt{5} + \sqrt{2}$.

Step 2: Expand the denominator.

$$\frac{6(\sqrt{5} + \sqrt{2})}{5 - 2} = \frac{6(\sqrt{5} + \sqrt{2})}{3}$$

$$(\sqrt{5})^2 - (\sqrt{2})^2 = 5 - 2 = 3.$$

Step 3: Simplify by dividing by 3.

$$2(\sqrt{5} + \sqrt{2})$$

$$\frac{6}{3} = 2. \text{ So } k = 2, x = 5, y = 2.$$

Final answer: $2(\sqrt{5} + \sqrt{2})$

Diagram.

$$\frac{\sqrt{5} + \sqrt{2}}{\sqrt{5} + \sqrt{2}} = 1 \text{ (conjugate)}$$

$$\frac{6}{\sqrt{5} - \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } \sqrt{5} + \sqrt{2}} 2(\sqrt{5} + \sqrt{2})$$

difference of two squares: $(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2}) = 5 - 2 = 3$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 33 (a6-033)

Difficulty: Foundation / Marks: 4

Express in the form $k(\sqrt{x} \pm \sqrt{y})$, where x and y are integers and k is an integer or fraction:

$$\frac{10}{\sqrt{7} + \sqrt{2}}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{7} - \sqrt{2}$.

$$\frac{10(\sqrt{7} - \sqrt{2})}{(\sqrt{7} + \sqrt{2})(\sqrt{7} - \sqrt{2})}$$

Switch the sign between the surds.

Step 2: Expand the denominator.

$$\frac{10(\sqrt{7} - \sqrt{2})}{7 - 2} = \frac{10(\sqrt{7} - \sqrt{2})}{5}$$

$$7 - 2 = 5.$$

Step 3: Simplify.

$$2(\sqrt{7} - \sqrt{2})$$

$$\frac{10}{5} = 2.$$

Final answer: $2(\sqrt{7} - \sqrt{2})$

Diagram.

$$\frac{\sqrt{7} - \sqrt{2}}{\sqrt{7} - \sqrt{2}} = 1 \quad (\text{conjugate})$$

$$\frac{10}{\sqrt{7} + \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } \sqrt{7} - \sqrt{2}} = 2(\sqrt{7} - \sqrt{2})$$

difference of two squares: $(\sqrt{7} + \sqrt{2})(\sqrt{7} - \sqrt{2}) = 7 - 2 = 5$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 34 (a6-034)

Difficulty: Foundation | Marks: 4

Express in the form $k(\sqrt{x} \pm \sqrt{y})$, where x and y are integers and k is an integer or fraction:
 $\frac{3}{\sqrt{10} - \sqrt{7}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{10} + \sqrt{7}$.

$$\frac{3(\sqrt{10} + \sqrt{7})}{(\sqrt{10} - \sqrt{7})(\sqrt{10} + \sqrt{7})}$$

Use the conjugate rule.

Step 2: Expand the denominator.

$$\frac{3(\sqrt{10} + \sqrt{7})}{10 - 7} = \frac{3(\sqrt{10} + \sqrt{7})}{3}$$

$$10 - 7 = 3.$$

Step 3: Cancel.

$$\sqrt{10} + \sqrt{7}$$

The 3s cancel, so $k = 1$, $x = 10$, $y = 7$.

Final answer: $\sqrt{10} + \sqrt{7}$

Diagram.

$$\frac{\sqrt{10} + \sqrt{7}}{\sqrt{10} + \sqrt{7}} = 1 \quad (\text{conjugate})$$

$$\frac{3}{\sqrt{10} - \sqrt{7}} \xrightarrow{\text{multiply by the conjugate } \sqrt{10} + \sqrt{7}} = \sqrt{10} + \sqrt{7}$$

difference of two squares: $(\sqrt{10} - \sqrt{7})(\sqrt{10} + \sqrt{7}) = 10 - 7 = 3$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 35 (a6-035)

Difficulty: Foundation / Marks: 4

Express in the form $k(\sqrt{x} \pm \sqrt{y})$, where x and y are integers and k is an integer or fraction:
 $\frac{8}{\sqrt{11} + \sqrt{3}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{11} - \sqrt{3}$.

$$\frac{8(\sqrt{11} - \sqrt{3})}{(\sqrt{11} + \sqrt{3})(\sqrt{11} - \sqrt{3})}$$

Flip the sign between the surds.

Step 2: Expand the denominator.

$$\frac{8(\sqrt{11} - \sqrt{3})}{11 - 3} = \frac{8(\sqrt{11} - \sqrt{3})}{8}$$

$$11 - 3 = 8.$$

Step 3: Cancel.

$$\sqrt{11} - \sqrt{3}$$

The 8s cancel.

Final answer: $\sqrt{11} - \sqrt{3}$

Diagram.

$$\frac{\sqrt{11} - \sqrt{3}}{\sqrt{11} - \sqrt{3}} = 1 \text{ (conjugate)}$$

$$\frac{8}{\sqrt{11} + \sqrt{3}} \xrightarrow[\text{multiply by the conjugate } \sqrt{11} - \sqrt{3}]{\text{difference of two squares: } (\sqrt{11} + \sqrt{3})(\sqrt{11} - \sqrt{3}) = 11 - 3 = 8}} \sqrt{11} - \sqrt{3}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 36 (a6-036)

Difficulty: Foundation / Marks: 2

Rationalise and simplify: $\frac{7}{\sqrt{11}}$

Worked solution.

Multiply the top and bottom by $\sqrt{11}$.

$$\frac{7\sqrt{11}}{11}$$

$$\sqrt{11} \times \sqrt{11} = 11.$$

Final answer: $\frac{7\sqrt{11}}{11}$

Diagram.

$$\frac{\sqrt{11}}{\sqrt{11}} = 1$$

$$\frac{7}{\sqrt{11}} \xrightarrow[\text{denominator: } \sqrt{11} \times \sqrt{11} = 11]{\text{multiply top and bottom by } \sqrt{11}} \frac{7\sqrt{11}}{11}$$

Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 37 (a6-037)

Difficulty: Foundation / Marks: 3

Rationalise and simplify: $\frac{\sqrt{5}}{\sqrt{12}}$

Worked solution.

Step 1: Simplify $\sqrt{12}$.

$$\frac{\sqrt{5}}{\sqrt{12}} = \frac{\sqrt{5}}{2\sqrt{3}}$$

$$\sqrt{12} = 2\sqrt{3}.$$

Step 2: Multiply the top and bottom by $\sqrt{3}$.

$$\frac{\sqrt{5} \times \sqrt{3}}{2\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{15}}{6}$$

$$\sqrt{5} \times \sqrt{3} = \sqrt{15} \text{ and } 2\sqrt{3} \times \sqrt{3} = 2 \times 3 = 6.$$

Final answer: $\frac{\sqrt{15}}{6}$

Diagram.

$$\frac{\sqrt{5}}{\sqrt{12}} \xrightarrow{\sqrt{12} = 2\sqrt{3}} \frac{\sqrt{5}}{2\sqrt{3}} \xrightarrow{\times \frac{\sqrt{3}}{\sqrt{3}}} \frac{\sqrt{15}}{6}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 38 (a6-038)

Difficulty: Foundation / Marks: 4

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{\sqrt{200}}{5} + \frac{30}{\sqrt{50}}$

Worked solution.*Step 1: Simplify the first fraction.*

$$\frac{\sqrt{200}}{5} = \frac{10\sqrt{2}}{5} = 2\sqrt{2}$$

 $\sqrt{200} = 10\sqrt{2}$, then divide by 5.*Step 2: Rationalise and simplify the second fraction.*

$$\frac{30}{\sqrt{50}} = \frac{30}{5\sqrt{2}} = \frac{6}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

 $\sqrt{50} = 5\sqrt{2}$, simplify to $\frac{6}{\sqrt{2}}$, then rationalise.*Step 3: Add the like surds.*

$$2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$$

$$2 + 3 = 5.$$

Final answer: $5\sqrt{2}$ **Diagram.**

$$\frac{\sqrt{200}}{5} + \frac{30}{\sqrt{50}} \xrightarrow{\text{simplify and rationalise}} 2\sqrt{2} + 3\sqrt{2} \xrightarrow{\text{collect}} 5\sqrt{2}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 39 (a6-039)*Difficulty: Foundation / Marks: 2***Rationalise the denominator:** $\frac{2}{3\sqrt{5}}$ **Worked solution.***Multiply the top and bottom by $\sqrt{5}$.*

$$\frac{2\sqrt{5}}{3 \times 5} = \frac{2\sqrt{5}}{15}$$

$$3\sqrt{5} \times \sqrt{5} = 3 \times 5 = 15.$$

Final answer: $\frac{2\sqrt{5}}{15}$ **Diagram.**

$$\frac{2}{3\sqrt{5}} \xrightarrow[\text{denominator: } 3\sqrt{5} \times \sqrt{5} = 15]{\begin{array}{c} \boxed{\frac{\sqrt{5}}{\sqrt{5}} = 1} \\ \text{multiply top and bottom by } \sqrt{5} \end{array}} = \frac{2\sqrt{5}}{15}$$

Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 40 (a6-040)

Difficulty: Foundation / Marks: 2

Rationalise the denominator and simplify: $\frac{\sqrt{45}}{\sqrt{5}}$

Worked solution.

Step 1: Combine the surds under a single root.

$$\sqrt{\frac{45}{5}} = \sqrt{9}$$

Use $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$.

Step 2: Evaluate.

3

$$\sqrt{9} = 3.$$

Final answer: 3

Diagram.

$$\frac{\sqrt{45}}{\sqrt{5}} \xrightarrow{\text{one root}} \sqrt{\frac{45}{5}} = \sqrt{9} \xrightarrow{\text{evaluate}} 3$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 41 (a6-041)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{6}{4-\sqrt{10}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $4 + \sqrt{10}$.

$$\frac{6(4 + \sqrt{10})}{(4 - \sqrt{10})(4 + \sqrt{10})}$$

Flip the sign to get the conjugate.

Step 2: Expand the denominator.

$$\frac{6(4 + \sqrt{10})}{16 - 10} = \frac{6(4 + \sqrt{10})}{6}$$

$$4^2 - 10 = 16 - 10 = 6.$$

Step 3: Cancel.

$$4 + \sqrt{10}$$

The 6s cancel exactly.

Final answer: $4 + \sqrt{10}$

Diagram.

$$\frac{4 + \sqrt{10}}{4 + \sqrt{10}} = 1 \text{ (conjugate)}$$

$$\frac{6}{4 - \sqrt{10}} \xrightarrow{\text{multiply by the conjugate } 4 + \sqrt{10}} = 4 + \sqrt{10}$$

difference of two squares: $(4 - \sqrt{10})(4 + \sqrt{10}) = 16 - 10 = 6$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 42 (a6-042)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{32}{7 - \sqrt{17}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $7 + \sqrt{17}$.

$$\frac{32(7 + \sqrt{17})}{(7 - \sqrt{17})(7 + \sqrt{17})}$$

Use the conjugate to rationalise.

Step 2: Expand the denominator.

$$\frac{32(7 + \sqrt{17})}{49 - 17} = \frac{32(7 + \sqrt{17})}{32}$$

$$49 - 17 = 32.$$

Step 3: Cancel the common factor of 32.

$$7 + \sqrt{17}$$

Numerator and denominator both contain 32, giving $a = 7$, $b = 1$, $k = 17$.

Final answer: $7 + \sqrt{17}$

Diagram.

$$\frac{7 + \sqrt{17}}{7 + \sqrt{17}} = 1 \text{ (conjugate)}$$

$$\frac{32}{7 - \sqrt{17}} \xrightarrow{\text{multiply by the conjugate } 7 + \sqrt{17}} = 7 + \sqrt{17}$$

difference of two squares: $(7 - \sqrt{17})(7 + \sqrt{17}) = 49 - 17 = 32$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 43 (a6-043)

Difficulty: Foundation / Marks: 3

Rationalise the denominator and simplify: $\frac{12}{\sqrt{8}}$

Worked solution.

Step 1: Simplify $\sqrt{8}$.

$$\frac{12}{\sqrt{8}} = \frac{12}{2\sqrt{2}} = \frac{6}{\sqrt{2}}$$

$\sqrt{8} = 2\sqrt{2}$, then cancel 2 on top and bottom.

Step 2: Multiply the top and bottom by $\sqrt{2}$.

$$\frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

$\sqrt{2} \times \sqrt{2} = 2$, then divide.

Final answer: $3\sqrt{2}$

Diagram.

$$\frac{12}{\sqrt{8}} \xrightarrow{\sqrt{8} = 2\sqrt{2}} \frac{12}{2\sqrt{2}} = \frac{6}{\sqrt{2}} \xrightarrow{\text{rationalise}} \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 44 (a6-044)

Difficulty: Foundation / Marks: 4

Solve the equation $10 = (\sqrt{3} - 1)x$, **giving your answer in the form** $a + b\sqrt{3}$ **where** a **and** b **are integers.**

Worked solution.

Step 1: Divide both sides by $\sqrt{3} - 1$.

$$x = \frac{10}{\sqrt{3} - 1}$$

Isolate x .

Step 2: Multiply top and bottom by the conjugate $\sqrt{3} + 1$.

$$x = \frac{10(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = \frac{10(\sqrt{3} + 1)}{3 - 1}$$

$$(\sqrt{3})^2 - 1 = 2.$$

Step 3: Simplify.

$$x = \frac{10(\sqrt{3} + 1)}{2} = 5(\sqrt{3} + 1) = 5 + 5\sqrt{3}$$

$\frac{10}{2} = 5$, then expand the bracket.

Final answer: $x = 5 + 5\sqrt{3}$

Diagram.

$$x = \frac{10}{\sqrt{3}-1} \xrightarrow{\times \text{ conjugate}} \frac{10(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{10(\sqrt{3}+1)}{2} \xrightarrow{\text{ simplify }} 5 + 5\sqrt{3}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 45 (a6-045)

Difficulty: Foundation / Marks: 5

Solve the equation $4 + \sqrt{5} = (2 - \sqrt{5})y$, **giving your answer in the form** $p + q\sqrt{5}$ **where** p **and** q **are integers.**

Worked solution.

Step 1: Divide both sides by $2 - \sqrt{5}$.

$$y = \frac{4 + \sqrt{5}}{2 - \sqrt{5}}$$

Isolate y .

Step 2: Multiply top and bottom by the conjugate $2 + \sqrt{5}$.

$$y = \frac{(4 + \sqrt{5})(2 + \sqrt{5})}{(2 - \sqrt{5})(2 + \sqrt{5})}$$

Use the conjugate to eliminate the surd from the denominator.

Step 3: Expand the numerator and denominator.

$$y = \frac{8 + 4\sqrt{5} + 2\sqrt{5} + 5}{4 - 5} = \frac{13 + 6\sqrt{5}}{-1}$$

Numerator: $8 + 6\sqrt{5} + 5 = 13 + 6\sqrt{5}$. Denominator: $4 - 5 = -1$.

Step 4: Simplify.

$$y = -13 - 6\sqrt{5}$$

Dividing by -1 negates each term.

Final answer: $y = -13 - 6\sqrt{5}$

Diagram.

$$y = \frac{4 + \sqrt{5}}{2 - \sqrt{5}} \xrightarrow{\times \text{ conjugate}} \frac{(4 + \sqrt{5})(2 + \sqrt{5})}{4 - 5} \xrightarrow{\text{ simplify }} \frac{13 + 6\sqrt{5}}{-1} = -13 - 6\sqrt{5}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 46 (a6-046)

Difficulty: Foundation / Marks: 5

A rectangle has an area of $(3 + \sqrt{5}) \text{ cm}^2$ and a width of $(\sqrt{5} - 1) \text{ cm}$. Find the length of the rectangle. Give your answer in the form $a + b\sqrt{5}$, where a and b are integers.

Worked solution.

Step 1: Use the formula $\text{length} = \frac{\text{area}}{\text{width}}$.

$$\text{length} = \frac{3 + \sqrt{5}}{\sqrt{5} - 1}$$

Divide the area by the width.

Step 2: Multiply top and bottom by the conjugate $\sqrt{5} + 1$.

$$\frac{(3 + \sqrt{5})(\sqrt{5} + 1)}{(\sqrt{5} - 1)(\sqrt{5} + 1)}$$

The conjugate eliminates the surd from the denominator.

Step 3: Expand the numerator and denominator.

$$\frac{3\sqrt{5} + 3 + 5 + \sqrt{5}}{5 - 1} = \frac{8 + 4\sqrt{5}}{4}$$

Numerator: $3\sqrt{5} + 3 + \sqrt{5} \times \sqrt{5} + \sqrt{5} = 3\sqrt{5} + 3 + 5 + \sqrt{5} = 8 + 4\sqrt{5}$. Denominator: $5 - 1 = 4$.

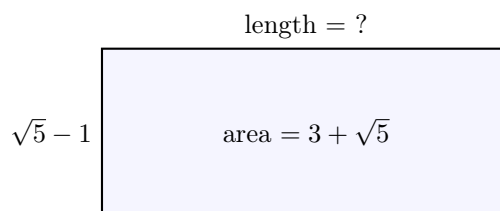
Step 4: Simplify.

$$2 + \sqrt{5}$$

Divide each term by 4.

Final answer: $(2 + \sqrt{5}) \text{ cm}$

Diagram.



length = area $\div$ width = $\frac{3 + \sqrt{5}}{\sqrt{5} - 1}$; multiplying by the conjugate $\frac{\sqrt{5} + 1}{\sqrt{5} + 1}$ gives $\frac{(3 + \sqrt{5})(\sqrt{5} + 1)}{4} = \frac{8 + 4\sqrt{5}}{4} = 2 + \sqrt{5} \text{ cm}$.

Rationalising the Denominator 47 (a6-047)

Difficulty: Foundation / Marks: 3

Rationalise the denominator: $\frac{1}{\sqrt{2} + \sqrt{3}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{2} - \sqrt{3}$.

$$\frac{\sqrt{2} - \sqrt{3}}{(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3})}$$

Use the conjugate to eliminate surds from the denominator.

Step 2: Expand the denominator.

$$\frac{\sqrt{2} - \sqrt{3}}{2 - 3} = \frac{\sqrt{2} - \sqrt{3}}{-1}$$

$$(\sqrt{2})^2 - (\sqrt{3})^2 = 2 - 3 = -1.$$

Step 3: Simplify by multiplying through by -1 .

$$\sqrt{3} - \sqrt{2}$$

Dividing by -1 flips the signs.

Final answer: $\sqrt{3} - \sqrt{2}$

Diagram.

$$\frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} - \sqrt{3}} = 1 \quad (\text{conjugate})$$

$$\frac{1}{\sqrt{2} + \sqrt{3}} \xrightarrow[\text{multiply by the conjugate } \sqrt{2} - \sqrt{3}]{\text{difference of two squares: } (\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) = 2 - 3 = -1}} \sqrt{3} - \sqrt{2}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 48 (a6-048)

Difficulty: Foundation / Marks: 4

Express in the form $p + q\sqrt{r}$ where r is an integer, and p and q are integers or fractions:

$$\frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} - \sqrt{2}}$$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{6} + \sqrt{2}$.

$$\frac{(\sqrt{6} + \sqrt{2})(\sqrt{6} + \sqrt{2})}{(\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2})}$$

Use the conjugate of the denominator.

Step 2: Expand the numerator and denominator.

$$\frac{6 + 2\sqrt{12} + 2}{6 - 2} = \frac{8 + 2\sqrt{12}}{4}$$

Numerator: $(\sqrt{6})^2 + 2\sqrt{6}\sqrt{2} + (\sqrt{2})^2 = 6 + 2\sqrt{12} + 2 = 8 + 2\sqrt{12}$. Denominator: $6 - 2 = 4$.

Step 3: Simplify $\sqrt{12} = 2\sqrt{3}$.

$$\frac{8 + 4\sqrt{3}}{4} = 2 + \sqrt{3}$$

Divide each term by 4.

Final answer: $2 + \sqrt{3}$

Diagram.

$$\frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} + \sqrt{2}} = 1 \quad (\text{conjugate})$$

$$\frac{\sqrt{6} + \sqrt{2}}{\sqrt{6} - \sqrt{2}} \xrightarrow{\text{multiply by the conjugate } \sqrt{6} + \sqrt{2}} = 2 + \sqrt{3}$$

difference of two squares: $(\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2}) = 6 - 2 = 4$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 49 (a6-049)

Difficulty: Foundation / Marks: 4

Rationalise the denominator: $\frac{\sqrt{2}}{3\sqrt{2}+1}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $3\sqrt{2} - 1$.

$$\frac{\sqrt{2}(3\sqrt{2} - 1)}{(3\sqrt{2} + 1)(3\sqrt{2} - 1)}$$

Flip the sign between the terms.

Step 2: Expand the numerator and denominator.

$$\frac{3 \times 2 - \sqrt{2}}{9 \times 2 - 1} = \frac{6 - \sqrt{2}}{17}$$

Numerator: $\sqrt{2} \times 3\sqrt{2} = 6$ and $\sqrt{2} \times (-1) = -\sqrt{2}$. Denominator: $(3\sqrt{2})^2 - 1 = 18 - 1 = 17$.

Final answer: $\frac{6 - \sqrt{2}}{17}$

Diagram.

$$\frac{3\sqrt{2} - 1}{3\sqrt{2} - 1} = 1 \quad (\text{conjugate})$$

$$\frac{\sqrt{2}}{3\sqrt{2} + 1} \xrightarrow{\text{multiply by the conjugate } 3\sqrt{2} - 1} = \frac{6 - \sqrt{2}}{17}$$

difference of two squares: $(3\sqrt{2} + 1)(3\sqrt{2} - 1) = 18 - 1 = 17$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 50 (a6-050)

Difficulty: Foundation / Marks: 2

Show that $\frac{12}{\sqrt{3}} = 4\sqrt{3}$.

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{3}$.

$$\frac{12}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{12\sqrt{3}}{3}$$

$$\sqrt{3} \times \sqrt{3} = 3.$$

Step 2: Simplify.

$$\frac{12\sqrt{3}}{3} = 4\sqrt{3}$$

$12 \div 3 = 4$, showing the result.

Final answer: $4\sqrt{3}(\text{shown})$

Diagram.

$$\frac{12}{\sqrt{3}} \xrightarrow{\times \frac{\sqrt{3}}{\sqrt{3}}} \frac{12\sqrt{3}}{3} \xrightarrow{\text{simplify}} 4\sqrt{3}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 51 (a6-051)

Difficulty: Foundation / Marks: 2

Rationalise and simplify: $\frac{\sqrt{27}}{\sqrt{3}}$

Worked solution.

Step 1: Combine under a single root.

$$\sqrt{\frac{27}{3}} = \sqrt{9}$$

Use $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$.

Step 2: Evaluate.

$$3$$

$$\sqrt{9} = 3.$$

Final answer: 3

Diagram.

$$\frac{\sqrt{27}}{\sqrt{3}} \xrightarrow{\text{one root}} \sqrt{\frac{27}{3}} = \sqrt{9} \xrightarrow{\text{evaluate}} 3$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 52 (a6-052)

Difficulty: Foundation / Marks: 4

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{11}{2\sqrt{3}-1}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $2\sqrt{3} + 1$.

$$\frac{11(2\sqrt{3} + 1)}{(2\sqrt{3} - 1)(2\sqrt{3} + 1)}$$

Use the conjugate to eliminate the surd.

Step 2: Expand the denominator.

$$\frac{11(2\sqrt{3} + 1)}{12 - 1} = \frac{11(2\sqrt{3} + 1)}{11}$$

$$(2\sqrt{3})^2 - 1 = 12 - 1 = 11.$$

Step 3: Cancel the common factor of 11.

$$2\sqrt{3} + 1 = 1 + 2\sqrt{3}$$

Giving $a = 1$, $b = 2$, $k = 3$.

Final answer: $1 + 2\sqrt{3}$

Diagram.

$$\frac{2\sqrt{3} + 1}{2\sqrt{3} + 1} = 1 \text{ (conjugate)}$$

$$\frac{11}{2\sqrt{3} - 1} \xrightarrow[\text{multiply by the conjugate } 2\sqrt{3} + 1]{\text{difference of two squares: } (2\sqrt{3} - 1)(2\sqrt{3} + 1) = 12 - 1 = 11} 1 + 2\sqrt{3}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 53 (a6-053)

Difficulty: Foundation / Marks: 4

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{11}{2\sqrt{5}+3}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $2\sqrt{5} - 3$.

$$\frac{11(2\sqrt{5} - 3)}{(2\sqrt{5} + 3)(2\sqrt{5} - 3)}$$

Flip the sign between the terms.

Step 2: Expand the denominator.

$$\frac{11(2\sqrt{5} - 3)}{20 - 9} = \frac{11(2\sqrt{5} - 3)}{11}$$

$$(2\sqrt{5})^2 - 3^2 = 20 - 9 = 11.$$

Step 3: Cancel the common factor of 11.

$$2\sqrt{5} - 3 = -3 + 2\sqrt{5}$$

Giving $a = -3$, $b = 2$, $k = 5$.

Final answer: $-3 + 2\sqrt{5}$

Diagram.

$$\frac{2\sqrt{5} - 3}{2\sqrt{5} - 3} = 1 \quad (\text{conjugate})$$

$$\frac{11}{2\sqrt{5} + 3} \xrightarrow[\text{multiply by the conjugate } 2\sqrt{5} - 3]{\text{difference of two squares: } (2\sqrt{5} + 3)(2\sqrt{5} - 3) = 20 - 9 = 11} -3 + 2\sqrt{5}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 54 (a6-054)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{25}{\sqrt{5}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{5}$.

$$\frac{25\sqrt{5}}{5}$$

$$\sqrt{5} \times \sqrt{5} = 5.$$

Step 2: Simplify.

$$5\sqrt{5}$$

$$25 \div 5 = 5.$$

Final answer: $5\sqrt{5}$

Diagram.

$$\frac{\sqrt{5}}{\sqrt{5}} = 1$$

$$\frac{25}{\sqrt{5}} \xrightarrow[\text{denominator: } \sqrt{5} \times \sqrt{5} = 5]{\text{multiply top and bottom by } \sqrt{5}} 5\sqrt{5}$$

Then $\frac{25\sqrt{5}}{5} = 5\sqrt{5}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 55 (a6-055)

Difficulty: Foundation / Marks: 2

Simplify, giving your answer in the form $p\sqrt{q}$ where p and q are integers: $\frac{18}{\sqrt{6}}$

Worked solution.

Step 1: Multiply the top and bottom by $\sqrt{6}$.

$$\frac{18\sqrt{6}}{6}$$

$$\sqrt{6} \times \sqrt{6} = 6.$$

Step 2: Simplify.

$$3\sqrt{6}$$

$$18 \div 6 = 3.$$

Final answer: $3\sqrt{6}$

Diagram.

$\frac{\sqrt{6}}{\sqrt{6}} = 1$

$$\frac{18}{\sqrt{6}} \xrightarrow[\text{denominator: } \sqrt{6} \times \sqrt{6} = 6]{\text{multiply top and bottom by } \sqrt{6}} = 3\sqrt{6}$$

Then $\frac{18\sqrt{6}}{6} = 3\sqrt{6}$. Multiply by $\frac{\sqrt{a}}{\sqrt{a}} = 1$: this leaves the value unchanged but clears the surd from the denominator.

Rationalising the Denominator 56 (a6-056)

Difficulty: Foundation / Marks: 2

Rationalise and simplify: $\frac{\sqrt{48}}{\sqrt{6}}$

Worked solution.

Step 1: Combine the surds under a single root.

$$\sqrt{\frac{48}{6}} = \sqrt{8}$$

Use the quotient rule for surds.

Step 2: Simplify $\sqrt{8}$.

$$\sqrt{4 \times 2} = 2\sqrt{2}$$

4 is the largest square factor of 8.

Final answer: $2\sqrt{2}$

Diagram.

$$\frac{\sqrt{48}}{\sqrt{6}} \xrightarrow{\text{one root}} \sqrt{\frac{48}{6}} = \sqrt{8} \xrightarrow{\text{simplify}} 2\sqrt{2}$$

Solution flow: each arrow applies one step.

Rationalising the Denominator 57 (a6-057)

Difficulty: Foundation / Marks: 3

Express in the form $a + b\sqrt{k}$ where a , b and k are integers: $\frac{2}{3-\sqrt{7}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $3 + \sqrt{7}$.

$$\frac{2(3 + \sqrt{7})}{(3 - \sqrt{7})(3 + \sqrt{7})}$$

Flip the sign between the terms.

Step 2: Expand the denominator.

$$\frac{2(3 + \sqrt{7})}{9 - 7} = \frac{2(3 + \sqrt{7})}{2}$$

$$9 - 7 = 2.$$

Step 3: Cancel.

$$3 + \sqrt{7}$$

The 2s cancel exactly.

Final answer: $3 + \sqrt{7}$

Diagram.

$$\frac{3 + \sqrt{7}}{3 + \sqrt{7}} = 1 \text{ (conjugate)}$$

$$\frac{2}{3 - \sqrt{7}} \xrightarrow[\text{multiply by the conjugate } 3 + \sqrt{7}]{\text{difference of two squares: } (3 - \sqrt{7})(3 + \sqrt{7}) = 9 - 7 = 2}} = 3 + \sqrt{7}$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 58 (a6-058)

Difficulty: Foundation / Marks: 4

Express in the form $k(\sqrt{x} \pm \sqrt{y})$, where x and y are integers and k is an integer or fraction:
 $\frac{15}{\sqrt{6} + \sqrt{3}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{6} - \sqrt{3}$.

$$\frac{15(\sqrt{6} - \sqrt{3})}{(\sqrt{6} + \sqrt{3})(\sqrt{6} - \sqrt{3})}$$

Switch the sign between the surds.

Step 2: Expand the denominator.

$$\frac{15(\sqrt{6} - \sqrt{3})}{6 - 3} = \frac{15(\sqrt{6} - \sqrt{3})}{3}$$

$$6 - 3 = 3.$$

Step 3: Simplify.

$$5(\sqrt{6} - \sqrt{3})$$

$$\frac{15}{3} = 5.$$

Final answer: $5(\sqrt{6} - \sqrt{3})$

Diagram.

$$\frac{\sqrt{6} - \sqrt{3}}{\sqrt{6} - \sqrt{3}} = 1 \text{ (conjugate)}$$

$$\frac{15}{\sqrt{6} + \sqrt{3}} \xrightarrow[\text{multiply by the conjugate } \sqrt{6} - \sqrt{3}]{\text{difference of two squares: } (\sqrt{6} + \sqrt{3})(\sqrt{6} - \sqrt{3}) = 6 - 3 = 3}} = 5(\sqrt{6} - \sqrt{3})$$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.

Rationalising the Denominator 59 (a6-059)

Difficulty: Foundation / Marks: 4

Rationalise the denominator: $\frac{\sqrt{3}}{\sqrt{15} - \sqrt{6}}$

Worked solution.

Step 1: Multiply top and bottom by the conjugate $\sqrt{15} + \sqrt{6}$.

$$\frac{\sqrt{3}(\sqrt{15} + \sqrt{6})}{(\sqrt{15} - \sqrt{6})(\sqrt{15} + \sqrt{6})}$$

Use the conjugate to eliminate surds from the denominator.

Step 2: Expand the denominator.

$$\frac{\sqrt{3}(\sqrt{15} + \sqrt{6})}{15 - 6} = \frac{\sqrt{3}(\sqrt{15} + \sqrt{6})}{9}$$

$$15 - 6 = 9.$$

Step 3: Expand the numerator using $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$.

$$\frac{\sqrt{45} + \sqrt{18}}{9} = \frac{3\sqrt{5} + 3\sqrt{2}}{9}$$

$$\sqrt{3} \times \sqrt{15} = \sqrt{45} = 3\sqrt{5} \text{ and } \sqrt{3} \times \sqrt{6} = \sqrt{18} = 3\sqrt{2}.$$

Step 4: Simplify by dividing by 3.

$$\frac{\sqrt{5} + \sqrt{2}}{3}$$

Divide each term in the numerator by 3.

Final answer: $\frac{\sqrt{5} + \sqrt{2}}{3}$

Diagram.

$$\frac{\sqrt{15} + \sqrt{6}}{\sqrt{15} + \sqrt{6}} = 1 \text{ (conjugate)}$$

$$\frac{\sqrt{3}}{\sqrt{15} - \sqrt{6}} \xrightarrow{\text{multiply by the conjugate } \sqrt{15} + \sqrt{6}} = \frac{\sqrt{5} + \sqrt{2}}{3}$$

difference of two squares: $(\sqrt{15} - \sqrt{6})(\sqrt{15} + \sqrt{6}) = 15 - 6 = 9$

Multiply by the conjugate over itself. The denominator is then a difference of two squares, $(A + B)(A - B) = A^2 - B^2$, so every surd cancels and the denominator becomes rational.
